

GanJiang: policy-governed XRD workflow

Self-Evolving

User request + constraints

XRD skill / confirmed plan

State, policy & capability checks

a Prepare artifacts

Local parsing + validation

Powder XRD

Metadata + optional CIFs

O Observed pattern

Full range / original scale

M Model input

10-90° / 3500 points
Nonnegative / max = 100

I Inspector input

Full range / headerless

b Recall candidates

Remote compute / conditional routing

Single-phase first in auto mode

User elements enable the second channel

XMatcher

Detect, then match

Merge & bound

IDs / phase sets
Source ranks
Review pool

XQueryer-LW

Needs user elements

XMatcher-AutoMix / XDecomposer

O input; explicit multi or auto fallback

c Verify hypotheses

Remote Inspector + local decision policy

Structure database and CIF export

One CIF per candidate / one set per mixture

Identification Inspector

I + CIF(s): trust, coverage & peak evidence
Unshifted baseline + eligible shift trial

Finalize & check evidence

Trust + elements + required peak gates
Top-3 + source-specific ranking + status
Publish after all candidates reach terminal state

User CIFs (direct)

CIFs + refinement authorization

Identification only

d

Refine, validate, and deliver

Prepare refinement

O + phase CIFs
Background correction
Confirmed settings

Rietveld engines

PyXplore / FullProf
GSAS-II
One job / eligible engines

Refinement Inspector

Observed vs calculated
Aligned backgrounds
Pass: reasonable = true

Select passing run

Inspector score
Then peak coverage
No pass: user review

Report & audit

Profiles + available CIFs
HTML / PNG / ZIPs
Execution logs + hashes

≤1 budget-only retry for allowed non-convergence

Single-phase rejection

Auto mode only

→ Data / evidence

- - - -> Conditional control

O / M / I: distinct pattern artifacts; labels denote each consumer input.